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Large-scale Probabilistic Forecasting of Consumer Engagement of CPG Products using Heterogeneous Web Data

Abdullah Al Imran ^{a,*}, Kasun S Perera ^b^a*Ai Palette Pte. Ltd., Singapore, Central Region, Singapore*^b*Ai Palette Pte. Ltd., Singapore, Central Region, Singapore*

Abstract

Consumer Packaged Goods (CPG) play a pivotal role in customer-centric industries. Understanding the distinctive features of such products and how customers engage with them is essential for CPG manufacturers to create customer-winning products. In this study, we explore an innovative data-centric approach to forecast customer engagement for CPG products by monitoring their digital evolution, particularly in the Snacks category in the USA. Traditional methods for consumer analysis such as surveys, focus groups are time consuming, costly and ineffective with limited scope. They also lack a forecasting component, making it difficult for CPG companies to make forward-looking decisions. However, with the emergence of big data, we can leverage user generated public data from social media, web search, e-commerce platforms to estimate consumer engagement and make long-term forecasts. To achieve this, we propose a systematic approach to accumulate and prepare large datasets from heterogeneous web sources for CPG products. We then use state-of-the-art deep learning based time series forecasting models to efficiently train and predict consumer engagement for the next 12 months, benchmarking their computational efficiency and forecasting performance. Our findings indicate that the DeepAR model outperforms all other models, with the lowest NRMSE (=0.378), RMSE (=14.848) MSE (=220.457), MASE (=0.871), and sMAPE (=0.306) values. Furthermore, we demonstrate methods for computing single forecasting points and prediction intervals using the forecasted sample distribution from the probabilistic models. The proposed approach will provide CPG businesses with valuable insights to make informed decisions about product development, marketing strategies, and supply chain management.

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E-mail addresses: imran@aipalette.com (Abdullah Al Imran ) , kasun@aipalette.com (Kasun S Perera ) .

1. Introduction

Consumer Packaged Goods (CPG) are an essential part of our regular lives and customer-centric industries, valued at 2.052 Trillion USD at the end of 2022, reaching 2.448 Trillion USD by 2028 [18]. As a result, understanding how consumers consume products and engage with them is critical for businesses to make informed decisions about product development, marketing strategies, supply chain management, and thriving in a competitive market. However, monitoring and understanding consumption patterns, and consumer engagement is a challenging real-world problem for CPG companies due to numerous obstacles in practice [6, 14].

The conventional and most prevalent approaches used by CPG companies to monitor and understand consumer engagement are surveys, focus groups, and competitive analysis [12, 20, 16]. While these methods can provide useful insights into consumer behavior, they can often be time consuming, cost intensive, and inefficient. All these approaches require large financial investments and human involvement in data collection, curation, analysis, and other physical resources. Furthermore, due to the limited scope of these methodologies, effectively measuring consumer engagement in a timely manner becomes problematic. On the other hand, due to the scale of CPG items, forecasting future trajectories of consumer engagement introduces an extra challenge that results in missed opportunities and wasted resources for businesses [4]. Given the current emergence of big data, an alternative data-centric approach is highly desirable among the CPG companies that can provide a systematic methodology for cost effective data collection, preparation and efficient large-scale forecasting of consumer engagement.

With the rapid growth of online and social media platforms, the availability of user-generated public data, such as social media posts and web searches, has increased significantly. Social media has created a sense of community around CPG products, specially food products. Individuals nowadays like sharing their eating habits and food reviews on social media since it allows them to easily exchange their culinary experiences with their friends and followers while also gaining social validation. We recognize that certain features of this user-generated content from social media, such as ‘likes’, ‘comments’, and ‘shares’ are key indicators of how people engage with a particular food product [13]. Similarly, searching for a particular product in the search engine expresses consumer intent to engage with that product. These features can be used together to measure the level of engagement and interest people have in a particular CPG product.

On the other hand, time-series forecasting for the fast moving consumer packaged goods has long been a challenging problem because of the volume, volatility, non-stationariness, and effect of external factors. Training a separate traditional forecasting model for each item in the market is not a technically scalable solution given the volume of CPG products. At the same time it is computationally expensive since most of the CPG products are fast moving and require frequent model retraining with updated data. Furthermore, making accurate forecasts for CPG products is an additional challenge given their volatility, non-stationariness, and added noise from external factors. Hence, it creates a trade-off while choosing a forecasting model to find a balance between scalability and accuracy of the predictions.

Considering the above mentioned scenarios, in this study, we propose an innovative approach by combining the user-generated heterogeneous data and deep learning based forecasting models for building a robust and scalable solution for the given problem. Our approach employs a systematic methodology for capturing user-generated public data from social media platforms such as Facebook, Instagram, and Twitter, as well as web search data from Google Trends, and thereafter transforming the data and deriving consumer engagement scores. To develop an efficient and scalable forecasting model, we experiment with five state-of-the-art (SOTA) deep learning-based forecasting models, including Encoder-Decoder LSTM, N-BEATS, DeepAR, Multi-headed Transformer, and Temporal Fusion Transformer, and benchmark their performance for this problem. We also show how to better optimize the hyperparameters of these models using Bayesian Optimization and how probabilistic deep learning based forecasting models can provide us more flexibility and confidence on the predictions. Our proposed approach in this work aims to fill the following limitations and research gaps within CPG domain by adding the following contributions:

1. Proposing a user-generated data ingestion and transformation approach from heterogeneous sources to estimate the consumer engagement of CPG products within a market.
2. Opening up a new frontier by providing a systematic methodology for efficient tuning, and training of SOTA deep learning models for long-term consumer engagement forecasting of large-scale CPG products.
3. Providing a performance benchmark for the SOTA deep learning based forecasting models and showing how we can achieve reliable and flexible predictions using probabilistic forecasting.

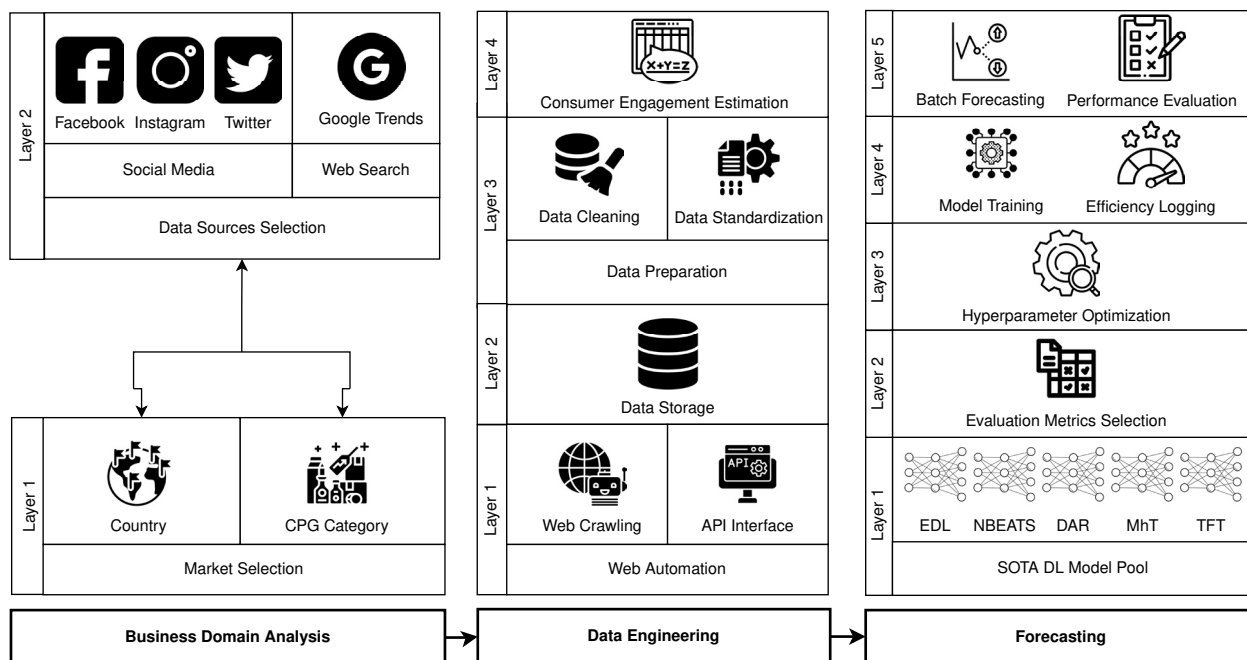


Fig. 1. Proposed Methodology

2. Background Study

Understanding consumer engagement and forecasting its future trajectory has long been a research focus for CPG companies. In this section, we'll provide a comprehensive study on the most frequently used approaches in the literature for consumer understanding, the paradigm shift from conventional methods to data and AI-driven approaches, and a few notable studies that have attempted to solve forecasting problems within the CPG domain.

Market research and competitive analysis have been the most traditional approaches to understand consumer behaviors and engagements. Market research [12] involves surveys, focus groups or other forms of direct feedback gathering, in which customers are asked to rate various aspects of the products. This approach typically consists of a collection of consumers who are invited to participate in an informal discussion about a specific product or market trend. During these sessions, participants are encouraged to share their opinions, insights, and experiences with the topic at hand. This information is then used to determine customer engagement, attitudes and preferences, as well as identify areas where companies could improve their own products. Competitive analysis, on the other hand, involves methodologies such as Customer Competitive Analysis (CCA), and Market Basket Analysis (MBA). CCA [20] involves identifying competitors who produce similar products and assessing how customers perceive these competitors' ability to meet their needs and requirements. MBA [16] entails analyzing the products that customers purchase together in a single transaction to identify patterns and trends in customer preferences.

With the advent of big data and AI, more recently, companies have shifted from traditional methods to data and AI-driven approaches to gain insights into customer behavior. Notably, retail giant Walmart [11] is renowned for its comprehensive data collection and analysis of information regarding purchasing history, and inventory movements. Through the analysis of historical transactional data, companies are developing dynamic customer segmentation [9] and profiling strategies that identify key customer groups based on their buying patterns and preferences. Additionally, by leveraging sophisticated machine learning algorithms and data mining tools, companies are now uncovering hidden relationships between seemingly unrelated products and use this information to drive upselling and cross-selling efforts [9]. Over the time, researchers have also noticed that customers are increasingly taking an individualized approach to their purchasing decisions [10]. With the rise of the internet and online information resources, customers have greater access to information about market offerings and can conduct research and evaluations to inform their decisions. To evaluate the usefulness of such user-generated data a study [23] was conducted considering Twitter as a

research tool, focusing on the topic of what people say when they tweet about different eating situations, rather than specific food and beverage products. The authors performed automated data collection and word analysis on 48,746 original tweets retrieved using the search words such as breakfast, lunch, dinner, and snack. Results from the study highlighted the significant potential of Twitter data for consumer behavior understanding and market research purposes. Another popular research area within this domain is sentiment analysis of food products from online consumer reviews. In [3], authors have developed a bot and tracked 568,454 food reviews of 74,258 products from 256,059 users on Amazon over a period of ten years and used NRC Emotion Lexicon to analyze the sentiment of the product reviews.

While exploring the scope of forecasting methods within this domain, we found most of the studies focused on addressing demand or sales forecasting problems. For instance, the author in [7] used Moving Average, Simple, Double, and Triple exponential methods for forecasting demands of agri-food products using the 5 years of historical sales data. In [21], the authors used ARIMA and exponential smoothing models to forecast the demand of ready-to-eat food products using weekly sales data of the previous 3 years. In [5], the authors used RBF Neural Network architecture in combination with genetic algorithms to forecast daily sales of dairy products using historical sales data of the previous 1 year. According to the survey paper [8] on demand forecasting methodologies, the most frequently used forecasting models are ARIMA, SARIMA, Exponential Smoothing, and Holt-Winters from traditional statistical family; Multiple, Poisson, Lasso Regression, Support Vector Machines, and XGBoost from machine learning family; Feed-Forward Neural Network, RNN, and LSTM from deep learning family.

To summarize the background study, we realized that most of the CPG companies have been using traditional market research and competitive analysis methods for understanding consumer behavior and engagement, while realizing the fact that such traditional approaches are time, cost, and resource intensive. Concerning that, companies started leveraging user-generated web data along with data mining techniques to derive consumer insights, but remained limited to social media posts and product review data. In terms of forecasting, a narrow scope has been explored within the demand or sales forecast problems using traditional statistical, machine learning, and deep learning approaches, while also remaining limited by developing univariate and separate models for each item to forecast. Our proposed approach aims to eradicate these gaps and enable businesses to become more data-driven.

3. Methodology

Our proposed methodology is divided into 3 phases: Business Domain Analysis, Data Engineering, and Forecasting. Each phase has different layers of operations. Business Domain Analysis has 2 layers including Market Selection and Data Source Selection. Data Engineering phase has 4 layers including Web Automation, Data Storage, Data Preparation, and Consumer Engagement Estimation. Finally, the 3rd and final phase, Forecasting has 5 layers including SOTA DL Model Pool, Evaluation Metrics Selection, Hyperparameter Optimization, Model Training & Performance Logging, and Batch Forecasting. The detailed workflow of our proposed approach is presented in Fig 1.

3.1. Business Domain Analysis

The first phase of the methodology starts with Business Domain Analysis where we choose our focus market and a list of data sources required for the later operations.

3.1.1. Market Selection

Market Selection includes choosing the country and the CPG category, depending on the business interest. While choosing a country and category, it is important to consider the maturity level of the market within that country and category. The more mature the market, the more data will be available for study. For this study we've chosen to go with the USA market. Since CPG encompasses a wide range of product categories such as food, beverages, household cleaners, toiletries, tobacco, and cosmetics, we have selected *snack* products within the food category as the focus of this study. Snacks are one of the most popular and diversified items in CPG food category, with a broad variety of tastes, textures, and packaging options, and they also have a considerable market share in CPG. However, our approach can be completely replicated to other CPG categories as well.

3.1.2. Data Sources Selection

It is the step in which we investigate relevant data sources to obtain user generated data that will provide us with the required attributes for estimating consumer engagement. In this study, we leverage two types of user-generated data sources: social media data and web search data. We are considering *Facebook*, *Instagram*, and *Twitter* as social media data sources since they are the most popular platforms, based on user penetration rates in the USA. Choosing which social media platforms to utilize is determined by the country's penetration of certain social media platforms. For example, if we select China as our market, the most preferred social media platforms are Weibo and TikTok. For web search data, we will capture search data from *Google Trends*. It may also vary depending on the country. In China, for example, Baidu is the most used search engine.

Algorithm 1: Consumer Engagement Estimation

Result: Total Consumer Engagement Score

product_list $\leftarrow$ from db;

facebook_data, *twitter_data*, *instagram_data*, *web_data* $\leftarrow$ from db;

social_data $\leftarrow$ None;

social_engagement_score $\leftarrow$ None;

total_engagement_score $\leftarrow$ None;

for *data* in (*facebook_data*, *twitter_data*, *instagram_data*) **do**

social_data.weighted_engagement_score $\leftarrow \alpha \times \text{likes} + \beta \times \text{comments} + (1.0 - \alpha - \beta) \times \text{shares}$;

end

social_engagement_score $\leftarrow$ *social_data.groupby([product_id, post_timestamp])*

.sum(weighted_engagement_score);

for *product_id* in *product_list* **do**

social_engagement_score[product_id, post_timestamp]

.scaled_engagement_score $\leftarrow$ *min_max_scaler*(

weighted_engagement_score, range = (0, 100));

end

for *product_id* in *product_list* **do**

total_engagement_score[product_id, post_timestamp] $\leftarrow$

web_data[product_id, timestamp].trend_value +

social_engagement_score[product_id, post_timestamp]

.scaled_engagement_score;

end

3.2. Data Engineering

Data Engineering is the second phase of our approach that involves data accumulation, storing and preparation activities. This phase is one of critical phases since better quality data results better analytics. In this phase, first we made a comprehensive list of snack products within the USA market. For, most of the CPG companies they have their internal data, however, in our case we prepared this list from e-commerce websites such as Amazon, eBay, Walmart. We've accumulated a list of 2600 products for this study.

3.2.1. Web Automation

Web Automation includes the automation process of collecting required data from the selected data sources for each of the products in the list. We've employed a keyword and hashtag based search approach to identify posts that has mentioned the product name. For, Facebook and Instagram we've developed a data collection bot to automate the process of doing keyword and hashtag based search for each of the products and collecting all the social posts. For, Twitter we've utilized their official API interface to do the same. For web search data collection, we've also used the official API of Google Trends. All the data collection processes ran as parallel jobs and returned data into JSON format. For this study, we're focusing to collect data on *monthly* frequency.

3.2.2. Data Storage

Data Storage involves the ingestion process of the data coming from web automation to a database. For our case, we have choose PostgreSQL as our primary database. We use 4 separate tables to store our social data and web search data. For social data, the attributes we store are 'product_id', 'product_name', 'post_timestamp', 'post_text', 'likes', 'comments', and 'shares'. For web search, we store 'product_id', 'product_name', 'timestamp', and 'trend_value'. In total we've collected 24,446,300 data points from social platforms and 2,83,400 data points from Google Trends.

3.2.3. Data Preparation

It includes steps for cleaning and standardization of the heterogeneous data that has been collected. Firstly, we formatted the 'timestamp' columns to monthly frequency. Secondly, for social media data points we identify the business or promotional posts and influencer or viral posts. These data points might introduce outliers to our data and skew the analysis of trend forecast. We identified promotional posts by using heuristics such as the most frequently used terms by businesses. Promotional content often includes a URL, phone number, or one or more phrases such as 'order now', 'call for orders', or 'DM to know more'. Thirdly, for influencer or viral post identification we've used z - score based statistical approach. Since, Instagram doesn't have share feature, we imputed the feature with 0.0. Then, for each product we calculated its z - score based on the 'likes', 'comments', and 'shares'. If a product has a z - score greater than 3 or less than -3 we considered those as outliers and excluded those data points from further analysis. For web search data, we imputed the missing values with 0.0. This process makes sure that each product has data for each month from 2014 Jan to 2023 Feb. The data preparation process of this research is largely facilitated by the sophisticated FOODIE framework proposed in [1].

3.2.4. Consumer Engagement Estimation

Once the cleaning and standardization is done, we leverage the data to estimate consumer engagements. We use algorithm 1 to estimate the *total_consumer_engagement_score*. First, we start by retrieving product list, social media data for Facebook, Twitter, Instagram, and web data from the database. Next, for each social media platform's data, we calculate the *weighted_engagement_score* for each post. While calculating the weighted engagement score we've used $\alpha = 0.2$ and $\beta = 0.4$ for Facebook and Twitter, $\alpha = 0.4$ and $\beta = 0.6$ for Instagram since Instagram doesn't has share. After calculating the weighted engagement score for each post, we group the *social_data* by *product_id* and *post_timestamp*, and sum up the weighted engagement score to get the *social_engagement_score*. Next, we loop through each *product_id* in the *product_list* and use the *min_max_scaler* to scale the weighted engagement score between 0 and 100. Finally, for each *product_id*, we calculate the *total_engagement_score* by combining the *scaled_engagement_score* with the web data *trend_value* at the corresponding *timestamp*.

3.3. Forecasting

Once the calculation of total consumer engagement score is done, we focus on the third phase of the methodology which is building forecasting models. It is evident from the background study that most of the approaches employed training individual statistical forecasting model for each item. Since it is technically not scalable, in the study, our approach focuses on training a single model for all the time series at once by leveraging SOTA deep learning algorithms. This approach will provide the following benefits:

- i Increased efficiency: With this approach it is possible to reduce the time and computational resources required for training, evaluating and optimizing multiple models for each individual item. This makes the process much more efficient and provide us flexibility for frequent retraining of the model, especially when dealing with large volume of time-serieses such as CPG products.
- ii Accurate forecast due to cross learning: This approach allows the model to better capture the interdependencies and relationships between the different time series, which may be missed when training individual models. Using cross learning can lead to more accurate and reliable forecasts.
- iii Improved generalization: This approach, when trained on a large volume of data, can better capture patterns and trends in the data, leading to improved generalization and performance on unseen data. This is especially true when dealing with new products or categories, where individual statistical forecasting models may be unavailable or unreliable.

- iv **Ease of productization:** This approach simplifies the productization and deployment process by reducing the complexity of managing multiple models, which can be time-consuming and error-prone. It also allows for real-time or near-real-time forecasting, which can be crucial for many business scenarios where quick decision-making is critical.

3.3.1. SOTA Deep Learning Model Pool

In this step we make a pool of State-of-The-Art Deep Learning based forecasting models for our use-case. While choosing the forecasting models, we have chosen two models such as Encoder-Decoder LSTM, and NBEATS from the non-probabilistic family, and three models such as DeepAR, Multi-headed Transformer, and Temporal Fusion Transformer from the probabilistic family. However, we use the non-probabilistic models as our baseline models.

Encoder-Decoder LSTM (EDL): EDL [24] is a deep learning forecasting model that uses a sequence-to-sequence architecture consisting of an encoder and decoder network with Long Short-Term Memory (LSTM) units. The encoder network processes the input sequence and extracts relevant features, while the decoder network generates the output sequence. EDL has several advantages over traditional models, including the ability to handle non-linear and non-stationary time series data, and the ability to capture long-term dependencies between input and output sequences.

Neural Basis Expansion Analysis for Time Series (NBEATS): NBEATS [17] is a deep learning forecasting model that uses a deep stack of MLPs with the doubly residual connections. It has a generic and interpretable architecture depending on the blocks it uses. NBEATS has several advantages over traditional models, including the ability to handle multiple seasonalities and long-term dependencies, and the ability to model complex patterns in the time series data.

DeepAR (DAR): DAR [19] is a probabilistic forecasting algorithm used in time-series analysis that is based on recurrent neural networks (RNNs). It is designed to model the dependencies among the time steps in a sequence and uses this information to make accurate future predictions. DeepAR consists of a multi-layered LSTM (Long Short-Term Memory) network that is trained to predict the future values of a time series, given its past values. The model uses an autoregressive formulation to predict the future values of a time series based on its past values. DAR has several advantages over traditional models, including the ability to handle missing values and the ability to incorporate external variables into the forecasting process.

Multi-headed Transformer (MhT): MhT [22] is a deep learning forecasting model based on the revolutionary Transformer architecture that uses a multi-headed attention mechanism to model time series data. It utilizes multiple attention heads to capture different aspects of the time series, such as seasonality, trends, and correlations. Each attention head attends to different positions in the input sequence and learns different representations of the time series, which are then concatenated and fed into a final fully connected layer for prediction. MhT has several advantages over traditional models, including the ability to handle multiple seasonalities and long-term dependencies, and the ability to model complex interactions between the input features.

Temporal Fusion Transformer (TFT): TFT [15] is a deep learning architecture that combines the strengths of the Transformer and the autoregressive methods for time-series forecasting. TFT processes the input time-series data using a multi-layer Transformer encoder, which is then concatenated with the decoded features obtained from a recurrent decoder. The TFT architecture utilizes the autoregressive property of time-series data by feeding the previous predictions as inputs to the decoder. Additionally, TFT integrates an exogenous input module that can handle external variables to improve forecasting accuracy. TFT is designed to handle complex time series with multiple seasonalities and long-term dependencies. TFT has several advantages over traditional models, including the ability to handle missing values and the ability to incorporate external variables into the forecasting process.

3.3.2. Evaluation Metrics Selection

In this step we chose a comprehensive list of evaluation metrics for assessing the performance of the forecasting models. Choosing a comprehensive list of performance metrics is necessary since different performance metrics provide different insights into the accuracy of the forecasting model, and using only one metric may not provide a complete picture of the model's performance. In this study, we have chosen 7 metrics such as Normalized Root Mean Squared Error (NRMSE), Root Mean Squared Error (RMSE), Mean Squared Error (MSE), Mean Absolute Scaled Error (MASE), Mean Absolute Percentage Error (MAPE), Symmetric Mean Absolute Percentage Error (sMAPE), and Mean Scaled Interval Score (MSIS). Since we're training a single model for multiple time-series, we will examine the overall aggregated performance of the models and create a performance benchmark.

3.3.3. Hyperparameter Optimization

For hyperparameter optimization of the models we've chosen a bayesian approach. Using a Bayesian approach [2] is preferable since it allows for a principled way of incorporating prior knowledge about the hyperparameters into the optimization process. This is particularly useful when the number of hyperparameters is large, and the search space is complex, as it helps to reduce the number of evaluations required to find the optimal hyperparameters. Table 1 presents the set of optimized hyperparameters.

Table 1. Optimized Hyperparameters

Hyperparameters	EDL	NBEATS	DAR	MhT	TFT
freq	M	M	M	M	M
context_length	24	24	24	24	24
prediction_length	12	12	12	12	12
learning_rate	0.001	0.001	0.001	0.001	0.001
batch_size	32	32	32	32	32
dropout_rate	-	-	0.1	0.1	0.1
attention_heads	-	-	-	8	4
patience	-	-	10	-	10
weight_decay	-	-	0.001	-	0.001
num_layers	2	2	2	-	-
hidden_dim	-	-	-	-	32
hidden_size	-	-	40	-	-
num_neurons	100	-	-	-	32

3.3.4. Model Training & Efficiency Logging

This step starts training the models and monitoring the losses, and runtime efficiency. Before starting model training, firstly, we've prepared the training and testing datasets. Each of the time-series has 110 months of data points from *Jan* – 2014 to *Feb* – 2023. The input steps or context length that we've chosen is 24 months and the output steps is 12 month, which means we're using previous 2 year's historical data as the context to forecast the values for the next 1 year. For, training purpose we've taken data from *Jan* – 2014 to *Feb* – 2020, and kept data from *Mar* – 2020 to *Feb* – 2023 as hold-out dataset for testing purpose. Secondly, we've updated the hyperparameters of the models to the optimum ones obtained from hyperparameter tuning. To train the models we've used a Virtual Machine of type 'n1 – standard – 4' from Google Cloud Platform (GCP) that has 4 vCPUs, and 15 GB RAM. We've trained all the models for 100 epochs. To accelerate model the training we can choose more powerful machines with GPUs. However, in this study, we focus on experimenting how efficient the model are in an standard unaccelerated setting. During the training we've closely monitored the loss per epoch and at the end we've logged the runtime of the models.

3.3.5. Batch Forecasting and Performance Evaluation

We've taken each of the trained models and used batch forecasting for making predictions on the test data. Batch forecasting is a technique used for large-scale time series forecasting, which involves generating forecasts for multiple time series simultaneously in batches, rather than generating forecasts for each time series individually. This approach is advantageous since it can significantly reduce the computational time required for generating forecasts, as it allows for parallel processing and efficient use of computing resources. From the test data we've used *Mar* – 2020 to *Feb* – 2022 as input month and made forecast for the period of *Mar* – 2022 to *Feb* – 2023. After, forecasting on the test data, we've run our custom evaluator to evaluate the forecasting performance of the models with the ground truth using the aforementioned evaluation metrics.

4. Result Analysis

In this section we'll do a comprehensive analysis of the run-time and forecasting performance of the trained models. We'll start with doing an analysis of the Training (T_R) and Forecasting (F_R) run-time performance followed by an in-depth analysis of the aggregated forecasting performance. Table 2 shows the run-time performance of the models.

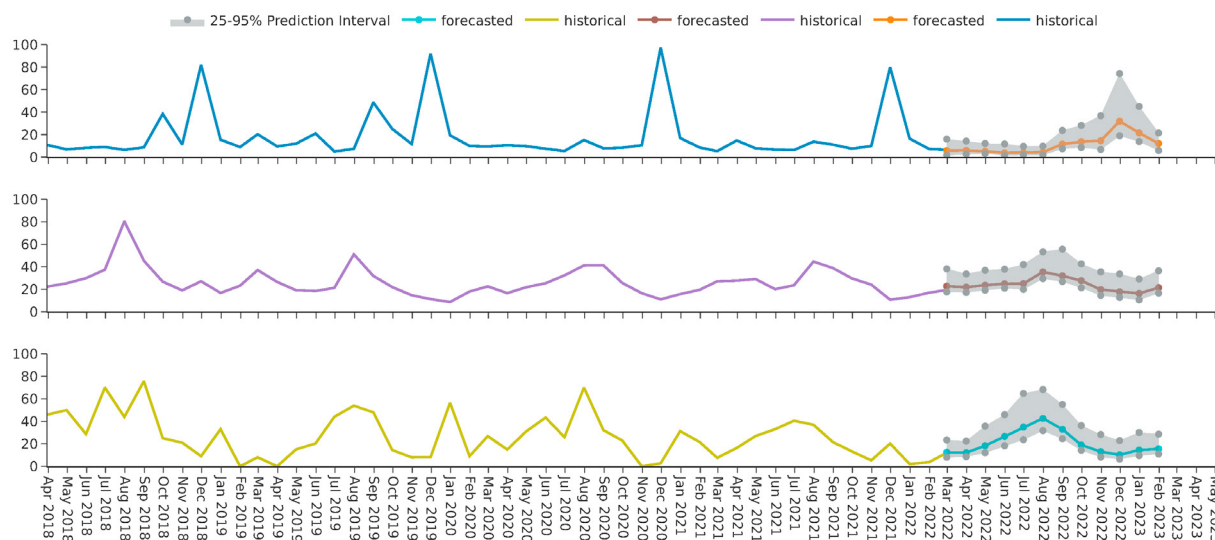


Fig. 2. Forecasts using DAR model with a 25-95% prediction interval

Table 2. Runtime performance of the models

	EDL	NBEATS	DAR	MhT	TFT
$T_R(sec.)$	309.71	586.46	429.22	515.39	563.19
$F_R(sec.)$	11.10	32.20	9.12	89.33	8.43

Table 3. Aggregated forecasting performance of the models

Metric	EDL	NBEATS	DAR	MhT	TFT
NRMSE	0.479	0.42	0.378	0.38	0.379
RMSE	18.8	16.485	14.848	14.91	14.894
MSE	353.45	271.77	220.46	222.32	221.83
MASE	1.057	0.893	0.871	0.895	0.86
MAPE	0.456	0.329	0.394	0.391	0.382
sMAPE	0.398	0.338	0.306	0.312	0.304
MSIS	42.275	35.724	7.329	6.65	6.023

Based on the above result, it is evident that the EDL model has the shortest training runtime of 309.71 seconds followed by DAR with 429.22 seconds, while the NBEATS model has the longest training runtime of 586.46 seconds. In terms of forecasting runtime, the TFT model has the shortest runtime of 8.43 seconds, while the MhT model has the longest runtime of 89.33 seconds.

The aggregated overall forecasting performance of the models on the test data is shown in Table 3. According to the results, EDL has the highest NRMSE value of 0.479 and the highest RMSE of 18.8, indicating that it performs the worst of the models. The MSE value of 353.45 is also relatively high compared to the other models, suggesting that it has higher forecasting errors. DAR has the lowest NRMSE value of 0.378 and lowest RMSE of 14.848, indicating that it has the best overall forecasting performance among the models. The MSE value of 220.46 is also relatively low in DAR, further supporting its good performance. It is also noticed that DAR, MhT, and TFT have very similar performance in terms of NRMSE, with values of 0.378, 0.38, and 0.379, respectively. Their RMSE values are also quite similar, ranging from 14.848 to 14.894. This suggests that these models are comparable in their forecasting accuracy. In terms of MASE, DAR has the lowest value of 0.871, NBEATS has the second-lowest MASE value of 0.893, indicating that these two models produces good predictions relative to the baseline. The other models have MASE values that are slightly higher, suggesting that they may be more susceptible to producing large errors. In terms

of MAPE and sMAPE, EDL has the highest values, suggesting that it may produce relatively larger errors compared to the other models. NBEATS, and DAR have the lowest MAPE and sMAPE values, indicating that these models can produce the most accurate percentage predictions. The MSIS values show that DAR, MhT, and TFT have significantly better performance than the other models, with values ranging from 6.023 to 7.329. NBEATS has a relatively higher MSIS value of 35.724, suggesting that it may not perform as well in forecasting uncertainty as the other models. Based on the overall evaluation metrics, NBEATS, DAR, MhT, and TFT have similar forecasting performance and outperform EDL. If we consider all the metrics together we can consider DAR as the best model since it outperforms all the models in NRMSE, RMSE, MSE, MASE, and sMAPE. However, the performance of these models can differ in terms of accuracy, percentage prediction, and forecasting uncertainty based on different scenarios and quality of the data. Hence, the choice of the best model depends on the specific requirements of the forecasting task.

5. Probabilistic Forecasting

Although we've considered EDL, and NBEATS models from non-probabilistic family, we emphasize using probabilistic models such as DAR, MhT, and TFT due to its outperforming performance and other advantages. Probabilistic forecasting models are becoming increasingly popular across different industries due to their robustness and prediction flexibility. Unlike traditional deterministic forecasting methods that provide a single point estimate, probabilistic forecasting models generate a range of possible outcomes and their associated probabilities. Models like DAR, MhT, and TFT makes probabilistic forecasts in the form of Monte Carlo samples and provides a representation of the underlying stochastic processes that drive the time-series. By accounting for the inherent variability and randomness in the data, probabilistic models can capture the full range of possible outcomes, including extreme events, which are often overlooked by traditional methods. In this study we've used equation 1 and 2 to transform the prediction samples into single forecasting points and derive a prediction interval for each month's prediction.

$$single_point_forecast_{mean} = \frac{1}{n} \sum_{i=1}^n x_{i,j} \quad (1)$$

where n is the number of rows, $x_{i,j}$ is the element in the i^{th} row and j^{th} column of the matrix, and the summation is taken over all rows for a fixed column j . The result after using this formula is a row vector of length 12, representing the column-wise means of the prediction sample. We can also chose median here based on different scenario. The following equation 2 has been used for calculating 25-95% prediction interval for all the time-series.

$$prediction_interval_{[upper,lower]} = \frac{(N+1) \cdot p}{100} \quad (2)$$

here, N represents the total number of points in the prediction sample, and p represents the desired percentile. When calculating the upper bound of the prediction interval we use $p = 95$, and for lower bound, we use $p = 25$. Such interval values provide us more confidence and flexibility while using the foretasted values in business decisions. Fig 2 presents a visualization of three randomly selected time series forecast using DAR model with a 25% – 95% prediction interval.

6. Limitations

It is important to recognize that this study is the first of its kind, introducing novel proposals for estimating consumer engagement for CPG products from heterogeneous web data and presenting data collection, ingestion and processing steps. However, the absence of a similar benchmark dataset posed a challenge when comparing the effectiveness of our models on other datasets. Also, the availability and quality of user-generated public data from social media, web search, and e-commerce platforms may vary depending on market and geography and may introduce potential biases. Despite these limitations, acknowledging and addressing them in future research will contribute to enhancing the effectiveness and applicability of data-centric forecasting approaches in the CPG industry.

7. Conclusion

This study demonstrated the potential of using user-generated web data and its significance in providing accurate monthly forecast of consumer engagement for CPG Snacks products, which can help businesses in the CPG industry to make informed decisions. By collecting heterogeneous data from multiple sources and preparing engagement scores for 2630 snack products, and utilizing 5 state-of-the-art time series forecasting models, the study proposed a systematic methodology for training single probabilistic forecasting models for all time-series. These models were then used to forecast the next 12 months' engagement scores using 24 months of historical context. The comparison of the forecasting performance and runtime efficiency revealed that DeepAR, Temporal Fusion Transformer, and Multi-headed Transformer were the best options.

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